From one bit to a fitted polynomial: engineered spectral filters for track reconstruction in a noisy LHCb VELO toy model

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Abstract

Track reconstruction in the LHCb vertex detector can be written as a linear system $A\mathbf{x} = \mathbf{b}$ over track *segments*, whose solution scores each segment by how strongly its neighbours reinforce it. Two earlier papers solved this system with quantum linear algebra: first with the HHL algorithm, and then far more cheaply with a one-bit variant, the one-bit quantum filter (1BQF), which replaces the full inversion by a single cosine filter of the Hamiltonian with one spectral zero. In this paper I develop that idea further. The 1BQF is the degree-one member of the family of polynomial spectral transforms provided by the quantum singular value transformation (QSVT), and once the problem is viewed this way the design question changes: it is no longer a question of which algorithm inverts A, but of which function of λ separates true from false segments. On clean events the answer is a four-line comb pinned to the eigenvalues of a five-hit track, and it drives the segment false rate from 20.5% (classical, T=1000 tracks) to 0.9% at 92% efficiency. On noisy events the answer must be measured: the false weight piles up on the eigenvalue lines of specific two- and three-segment motifs, and a polynomial fitted with roots at the measured pile-up beats the classical baseline at every attainable efficiency below a sharply defined *fragment floor*, the small population of true fragments that are spectral twins of the false motifs and that, on that Hamiltonian, no function of λ can separate. I also show that the two natural Hamiltonian modifications inherited from the classical literature can only be judged together with the response applied to them. The Denby–Peterson occupancy term destroys the one-bit filter, whose single zero cannot follow the spectrum it creates; but paired with a response refitted to that spectrum it becomes the strongest configuration in the study, roughly halving the fitted false rate at every matched efficiency, and, at small coupling, it breaks the spectral-twin degeneracy behind the fragment floor: the floor is a property of the base Hamiltonian, not of the problem. The bifurcation (fork) term behaves the same way, a regression under the clean-event comb and a net win under the refitted response. The result is a two-part design rule: the operator shapes the spectrum, and the fitted response separates it; neither knob can be evaluated without the other.

1 Introduction

Charged-particle track reconstruction is, at heart, a combinatorial matching problem: a detector hands us a cloud of hits, and we must decide which hits belong together. At the High-Luminosity LHC this cloud becomes dense enough that the matching cost dominates real-time reconstruction budgets, and the tracking community has responded by testing essentially every alternative

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computing platform: GPUs, FPGAs, neural networks, and, in the line of work this paper continues, quantum computers.

The lineage of this work is short and worth stating precisely. Nicotra et al. [1] showed that track reconstruction in a simplified model of the LHCb vertex detector (VELO) can be phrased as a linear system $A\mathbf{x} = \mathbf{b}$ over track segments and solved with the Harrow–Hassidim–Lloyd algorithm [2]; a restricted-phase-estimation variant of that construction was subsequently reported in ref. [4]. The follow-up paper [3] observed that the full inversion is overkill. The interesting information is which segments are active, not their precise weights, so a single-ancilla circuit, the one-bit quantum filter (1BQF), suffices. The 1BQF applies one cosine function of the Hamiltonian, placed so that its zero lands exactly on the eigenvalue shared by every isolated fake segment, and reads out the survivors. It is cheap, it suits near-term hardware, and it was demonstrated on trapped-ion and superconducting devices.

This paper asks the question that follows naturally from the 1BQF. A cosine with one zero is a polynomial filter of degree one[1]. The quantum singular value transformation (QSVT) [7, 10] tells us that a quantum computer can apply essentially any bounded polynomial of the Hamiltonian at a cost linear in its degree. So, what could we do with more than one zero? Which polynomial should we choose? And what happens to that choice when the Hamiltonian itself changes, because the detector is noisy or because we have added constraint terms to help it? In my opinion this last question carries most of the physics, and most of this paper is spent answering it.

I answer these questions on the same toy model as the two predecessor papers, extended with realistic noise. The answers can be summarised as a single working method, which I will refer to as fitting the response to the spectrum:

1. On clean events the true-track spectrum is *discrete*: detector geometry pins every true track to the same five-hit chain, whose four eigenvalues are known in closed form. A comb of narrow passes at those four lines, and nothing else, separates true from false essentially perfectly up to the densities I can simulate ($T = 1000$ tracks, four million segments), where the classical solver's false rate has climbed to 20.5%.

2. On noisy events the false population moves: it concentrates on the eigenvalue lines of specific small *motifs* (hit-sharing pairs and triples), which sit off the true lines but inside any naive pass window. The correct response cannot be written down a priori; it must be fitted to the measured pile-up. Once it is, the quantum false rate drops below the classical one at every efficiency below a floor that I can identify, count, and explain segment by segment.

3. Hamiltonian engineering and response engineering are coupled, and neither knob can be evaluated without the other. Both Denby–Peterson penalties destroy the one-bit filter and read as regressions under the fixed clean-event comb; both become net wins once the polynomial is refitted to the spectrum they produce. The occupancy term is the sharpest case: it breaks the notch outright, yet with a refitted response it roughly halves the false rate at every matched efficiency, and at small coupling it lifts the fragment floor by breaking the spectral-twin degeneracy that causes it. Changing the operator moves the activations, and moving the activations changes the polynomial we have to fit.

The paper is organised as follows. Section 2 builds the toy model and the segment Hamiltonian from zero, including the exact activation levels that make everything else quantitative. Section 3 reviews the 1BQF as a spectral filter and establishes the working-point conventions used throughout. Section 4 develops the QSVT machinery (block encoding, qubitization, and the linear-combination-of-Chebyshev construction) at the level of detail needed to count qubits and gates. Section 5 designs and evaluates the line comb on clean events, including its resource scaling

[1] Strictly, a trigonometric function realised at degree one in the Chebyshev basis of the walk operator; section 4 makes this exact.

and its provable limits. Section 6 introduces the noise model and shows what it does to each solver. Section 7 dissects the occupancy term's collision with the notch. Section 8 introduces the fork term and the design rule that every operator change demands a response refit. Section 9 fits the polynomial to the measured false pile-up and presents the main comparison, including the refitted occupancy result. Section 10 collects the limitations: realizability, generalization, Trotterization, and what a real detector geometry changes. Section 11 concludes.

A remark on style before we begin. This paper is deliberately verbose. The predecessor papers compressed the segment-Hamiltonian formalism to a few paragraphs; here I want every level, every eigenvalue, and every convention on the table, because the physics of why filters succeed and fail lives in exactly those details.

2 The toy model and the segment Hamiltonian

2.1 Events

The toy detector is the standard simplified VELO of refs. [1, 3]: parallel planes perpendicular to the beam axis (z), each recording a two-dimensional point where a charged particle crosses it. An event is generated by drawing T tracks from a common primary vertex on the z-axis, with polar angles inside a generation cone, and propagating each as a straight line through the planes. Every track leaves one hit per plane. Detector effects enter as controlled perturbations, and I will always state the triple in force: multiple-scattering kinks of width σ_{scatt} applied at each plane crossing, Gaussian hit-position smearing of width σ_{res}, and a hit-drop (inefficiency) probability applied hit by hit. The clean configuration used in section 5 is $(\sigma_{\text{scatt}}, \sigma_{\text{res}}, \text{drop}) = (10^{-4}, 0, 0)$; the heavy configuration used from section 6 onward is $(10^{-4}, 20\,\mu\text{m}, 1\%)$.

A *segment* is an ordered pair of hits on adjacent planes. With five planes and T hits per plane, each of the four inter-plane gaps offers T^2 candidate segments, so

$$n_{\text{seg}} = 4T^2, \qquad n_{\text{true}} = 4T, \qquad n_{\text{false}} = 4T(T-1), \tag{1}$$

where n_{true} counts the segments that connect two hits of the same generated track. The problem is a needle in a haystack, and the ratio $n_{\text{true}}/n_{\text{seg}} = 1/T$ says how deep the hay is: at $T = 1000$ only one segment in a thousand is real, and the false population grows quadratically while the true one grows linearly. Any segment classifier, classical or quantum, fights this composition before it fights anything else.

2.2 The Hamiltonian

Following Denby and Peterson's neural formulation of track finding [5, 6] as adapted by refs. [1, 3], we encode an event in a quadratic energy over segment activations $x_i \in \mathbb{R}$,

$$E(\mathbf{x}) = \tfrac{1}{2}\,\mathbf{x}^{\top} A\,\mathbf{x} - \delta\,\mathbf{1}^{\top}\mathbf{x}, \qquad A = (\gamma + \delta)\,I - C, \tag{2}$$

where C is the 0/1 *compatibility adjacency*: $C_{ij} = 1$ exactly when segments i and j share a hit as head-to-tail continuations (the end hit of one is the start hit of the other) and their mutual angle is below an acceptance ε. The constants are a self-penalty γ (discouraging indiscriminate activation) and a bias δ (rewarding activation); throughout the main text $\gamma = 3$, $\delta = 1$, so the constant diagonal is $s \equiv \gamma + \delta = 4$. Minimising eq. (2) gives the linear system

$$A\,\mathbf{x} = \delta\,\mathbf{1}, \tag{3}$$

which I solve classically with MINRES (A is symmetric, sparse, and at $\gamma = 3$ comfortably positive definite). A useful picture is a voting market: each segment starts with the same bias-funded credit δ, and the off-diagonal couplings let aligned neighbours reinforce one another. A segment embedded in a long, straight chain of compatible neighbours accumulates support; a segment with no compatible continuation is left with only its own bias against its own penalty.

The acceptance ε deserves its own paragraph, because it is the single most consequential experimental knob. It is set by a closed-form formula from the expected root-mean-square kink between consecutive true segments under the noise triple in force, with a 3σ margin so that true continuations pass the angular cut:

$$\varepsilon = \sqrt{2\,(3\sigma_{\mathrm{scatt}})^2 + 12\,\arctan^2(3\sigma_{\mathrm{res}}/\Delta z) + 2\,\theta_{\mathrm{min}}^2}\,, \tag{4}$$

where $\Delta z = 33\,\mathrm{mm}$ is the inter-plane spacing, the first term budgets the multiple-scattering kink between the two segments, the second the angle that hit-position smearing subtends across one plane gap, and $\theta_{\mathrm{min}} = 15\,\mu\mathrm{rad}$ is a numerical floor for the noise-free limit. On the heavy configuration eq. (4) gives $\varepsilon = 6.3\,\mathrm{mrad}$, fifteen times wider than the clean-configuration value, and section 6 shows what that width admits.

2.3 Exact activation levels: the atlas

Because A has constant diagonal, its action decomposes over the connected components of the compatibility graph, and small components can be solved by hand. This is worth doing once in full, because these little closed forms are the vocabulary of every result in the paper.

An *isolated* segment (no compatible neighbour) has the scalar equation $(\gamma + \delta)x = \delta$, so

$$x_{\mathrm{iso}} = \frac{\delta}{\gamma + \delta} = \frac{1}{4} = 0.250. \tag{5}$$

This is the false *attractor*: at any multiplicity, the overwhelming majority of the $4T(T-1)$ false segments are isolated and sit at exactly 0.250. A true track, by contrast, is a path P_4 of four segments (five hits), whose matrix is $sI - P_4$ with P_4 the path adjacency. Its solution can be written down exactly:

$$x_{\mathrm{outer}} = \frac{4}{11} \approx 0.364, \qquad x_{\mathrm{inner}} = \frac{5}{11} \approx 0.455, \tag{6}$$

where the outer levels belong to the two end segments (one continuation neighbour each) and the inner levels to the two middle segments (two neighbours each). More generally, a chain of m segments has outer activation

$$x_{\mathrm{outer}}(m) = \frac{1}{2} - \frac{r + r^m}{2\,(1 + r^{m+1})}, \qquad r = 2 - \sqrt{3}, \tag{7}$$

which climbs the ladder $\{0.250,\ \frac{1}{3},\ \frac{5}{14},\ \frac{4}{11}, \dots\} \nearrow 0.366$ for $m = 1, 2, 3, 4, \dots$ It is worth working two rungs by hand. A two-hit fragment (m=1) sits at exactly the isolated-false attractor 0.250, and a three-hit fragment (m=2) sits at 1/3, only 0.083 above it. Fragments are barely distinguishable from noise by amplitude alone; this innocent-looking fact returns as the fragment floor in section 9.

The classical decision rule is a threshold: segment i is *active* iff $x_i > \tau$. The *classical* convention, inherited from ref. [3] and kept for comparability, fixes

$$\tau(\gamma, \delta) = \frac{\delta}{\gamma + \delta} + 0.10, \tag{8}$$

i.e. $\tau = 0.35$ at $\gamma = 3$: a fixed margin above the false attractor, below the lowest clean-track level 4/11. It is important to say at the outset what this convention is and is not. For the classical solver it is a natural operating point, because the classical amplitude atlas happens to leave a gap

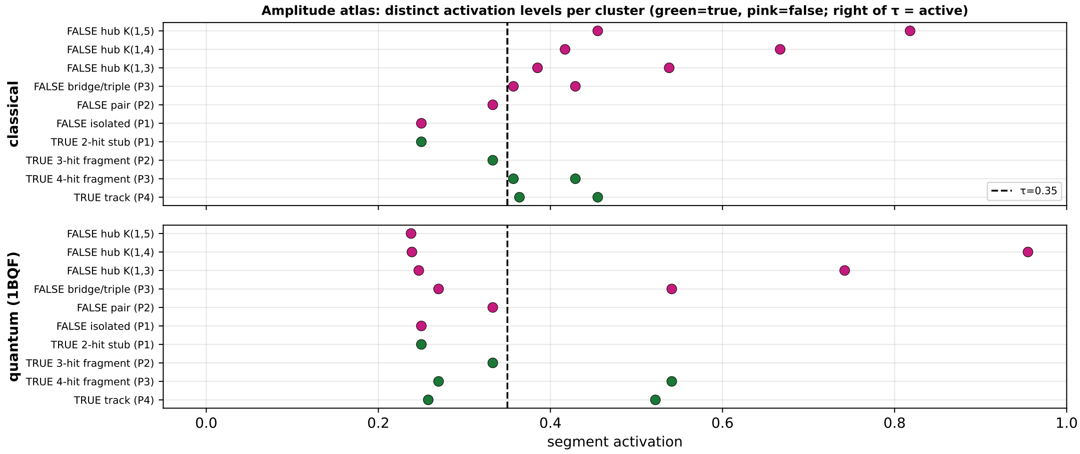


Figure 1: The amplitude atlas: exact activation level of every canonical cluster type (rows), classical (top) and 1BQF-filtered (bottom), with the fixed threshold $\tau = 0.35$ marked. Green: true; pink: false. Everything quantitative in this paper is a statement about where these levels sit relative to a threshold.

there. For a *filtered* solver it is in general the wrong place to read the device: a filter reshapes the amplitude atlas, and a threshold inherited from the unfiltered atlas can land in the middle of a reshaped true band. Every threshold choice trades efficiency against false rate, and the convention this paper actually uses for filtered solvers is *efficiency-first*: put the threshold where the wanted fraction of true segments survives, and quote the false rate that results. Section 3 makes this concrete, with the trade-off shown explicitly in figure 4.

2.4 Metric definitions

Every number in this paper is one of the following, and I will never use these words in any other sense:

- **segment efficiency** $\varepsilon_{\text{seg}} = n_{\text{true active}}/n_{\text{true}}$: the fraction of true segments that survive selection;
- **segment false rate** $f_{\text{seg}} = n_{\text{false active}}/n_{\text{active}}$: the fraction of the *selected* sample that is fake (note the denominator: this is an impurity, not a false-positive rate over all fakes);
- **working points**: either the *fixed threshold* of eq. (8); or the *efficiency-first working point* (the threshold placed just below the 1% quantile of the true-segment amplitudes, so $\varepsilon_{\text{seg}} \geq 0.99$ by construction — whenever this paper says "efficiency-first" without qualification, this 99% point is meant); or, in sections 8–9, the *matched-efficiency* convention: τ just below the k-th largest true amplitude with $k = \lceil \text{eff} \cdot n_{\text{true}} \rceil$, which lets solvers be compared at identical efficiency without quantile-grid artefacts.
- **AUC**, the threshold-free separation metric: sweeping τ from high to low traces out a curve of efficiency against false rate for each solver, and the *area under that curve* summarises how well the solver's amplitude ordering separates true from false segments independently of any particular threshold choice. AUC $= 1$ means some threshold separates them perfectly; AUC $= 0.5$ means the ordering is no better than chance. It is used when a single number for "separation quality" is needed, notably in section 7.

Quantum solutions are unit-norm vectors; before thresholding they are rescaled to the classical amplitude scale on the classically-active support. Rescaling by the full norm would be multiplicity-dependent, since the false bulk owns most of the norm, and was the source of a metric artefact documented in the programme's earlier write-ups. All thresholds act on $|x_i|$.

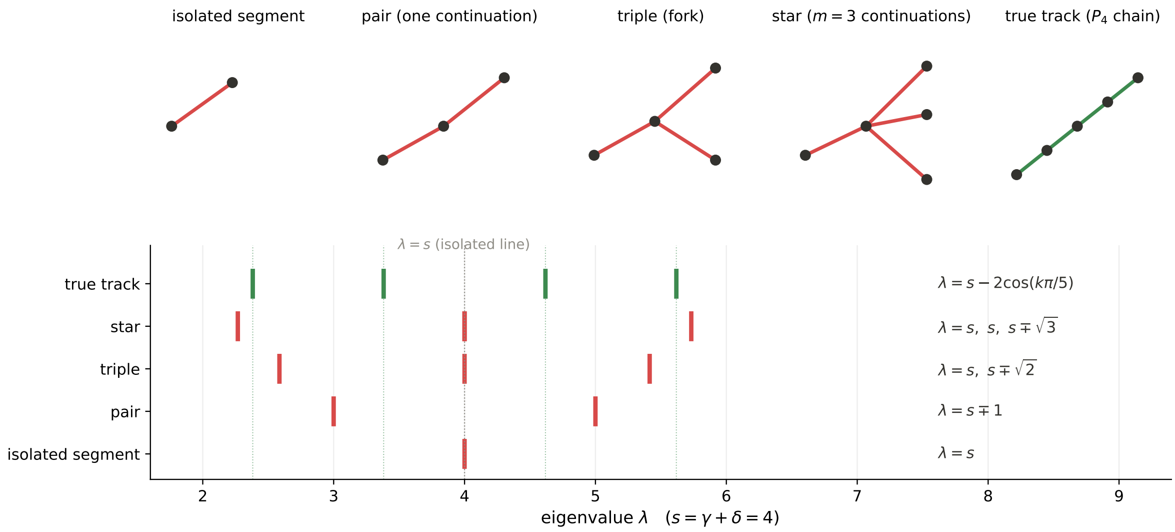


Figure 2: The motif dictionary. Top: the canonical segment clusters drawn in hit space (dots are hits, lines are segments; green true, red false). Bottom: the exact eigenvalue lines each motif contributes, from the closed forms of section 2.5, at $s = \gamma + \delta = 4$. Every false family sits close to, but never on, the four true-track lines (green dotted) — except for the spectral twins of section 5.5.

2.5 The spectrum

Because $A = sI - C$, its spectrum is a rigid reflection of the compatibility graph's: $\lambda(A) = s - \mu(C)$, where $\mu(C)$ denotes an eigenvalue of C, the ε-windowed 0/1 compatibility adjacency of section 2.2 ($C_{ij} = 1$ exactly when segments i and j continue one another head-to-tail within the angular acceptance ε, and 0 otherwise). Every spectral statement in this paper is therefore a statement about the eigenvalues of small 0/1 graphs. Three families matter:

1. **Isolated segments**: $\lambda = s$ exactly. This single line carries $\mathcal{O}(n_{\text{seg}})$ modes: at $T = 200$ under heavy noise, 155,371 of 157,609 segments (rep 0).
2. **Chains** P_m: $\lambda_k = s - 2\cos(k\pi/(m+1))$, $k = 1, \ldots, m$. A true track ($m = 4$) therefore lives on exactly four lines,
$$\lambda \in \{s - 2\cos(k\pi/5)\}_{k=1}^{4} = \{2.382,\ 3.382,\ 4.618,\ 5.618\} \quad (s = 4), \tag{9}$$
symmetric about s and split by the golden-ratio cosines[2].
3. **Stars** $K_{1,m}$ (one hit shared by m segments of the same role): $\lambda = s \pm \sqrt{m}$ plus $m-1$ copies of s.

Figure 2 draws each of these motifs in hit space and places its exact eigenvalue lines on a common λ-axis: this is the dictionary between what a cluster of segments *looks like* in the detector and where its solution weight *piles up* in the spectrum. The pair motif at $\lambda = s \pm 1$ and the triple at $\{s,\ s \pm \sqrt{2}\}$ do most of the false-population work under noise, and their lines interleave the true lines of eq. (9) without touching them. That gap, false lines close to but not on the true lines, is what this paper exploits. When noise is switched on in section 6, the *measured* spectral pile-up of a heavy-noise event (figure 10) lands exactly on the lines catalogued here. Motifs whose spectrum coincides with a true motif's are the exception, and they return in sections 5.5 and 9 as the limiting factor.

[2]The values $2\cos(\pi/5) = \varphi = 1.618\ldots$ and $2\cos(2\pi/5) = 1/\varphi$: the five-hit track is secretly a pentagon problem.

3 The one-bit quantum filter, read as a spectral filter

3.1 The circuit and its response function

The 1BQF of ref. [3] is a one-ancilla interference experiment. Prepare the uniform superposition $|u\rangle$ over the (zero-padded) segment register, put the ancilla in $(|0\rangle + |1\rangle)/\sqrt{2}$, apply e^{iAt} to the register controlled on the ancilla, close the interferometer with a second Hadamard, and keep the runs in which the ancilla reports the interference branch. The kept state is

$$|\psi\rangle \;\propto\; \tfrac{1}{2}(I + e^{iAt})|u\rangle \;=\; \sum_i e^{i\lambda_i t/2} \cos\Big(\frac{\lambda_i t}{2}\Big)\, \langle v_i|u\rangle\, |v_i\rangle, \tag{10}$$

where $\{\lambda_i, |v_i\rangle\}$ are the eigenpairs of A. Reading out $|\langle j|\psi\rangle|$ segment by segment gives amplitudes x_j^Q, and the whole device is summarised by a single real response function applied to the spectrum:

$$f(\lambda) = \cos\Big(\frac{\lambda t}{2}\Big), \qquad t = \frac{\pi}{s}. \tag{11}$$

The choice $t = \pi/s$ places the zero of the cosine exactly at $\lambda = s$, the isolated-segment line. Every isolated false segment, all $\mathcal{O}(n_{\mathrm{seg}})$ of them, is annihilated exactly: not suppressed, killed. This is the entire content of the 1BQF, and the form in which we will need it later is a simple accounting rule: each application of the filter buys exactly one zero of the response, and the 1BQF spends its one zero on the largest false population in the problem. The padding modes introduced to reach a power-of-two dimension are diagonal-only, sit at $\lambda = s$ too, and are removed by the same zero, a small but pleasing piece of engineering hygiene.

Figure 3 makes the geometry of this choice, and of everything that follows, visible. The top panel is the 1BQF response of eq. (11) drawn over the populated spectrum: its single zero sits exactly on the isolated-false line, and every other population, the pair and triple motif lines included, lands on the slopes of the cosine and passes with large weight. The filter attacks the isolated segments and nothing else; whatever false weight is *coupled* sails through. The bottom panel shows the QSVT alternative developed from section 4 onward: a degree-40 polynomial with narrow passes at the four true-track lines and effective zeros everywhere else, including the isolated, pair and triple lines simultaneously. The one-sentence summary of this paper is the difference between those two panels: a degree-one response owns one zero, a degree-d response owns d, and the noisy-event false spectrum demands more than one.

3.2 What the notch cannot do

Away from its zero the cosine is not flat, and this has two consequences that define the 1BQF's operating envelope.

The first is that true amplitudes are reshaped. A true track's four modes (eq. (9)) sit symmetrically about s, where $|f(\lambda)|$ is small; the filtered chain no longer reproduces the classical $\{4/11, 5/11\}$ profile. The effect has a closed form: the outer (endpoint) segments of a clean track drop from 0.364 to 0.258, below the fixed threshold $\tau = 0.35$, while the inner pair survives above it. At the fixed threshold the 1BQF therefore reports a segment efficiency plateau of ≈ 0.73–0.75: it finds the middles of tracks and loses the ends. This endpoint loss is a property of the cut, not of the physics; the endpoint amplitudes remain perfectly well ordered above the false ones, so a threshold placed below the endpoint band recovers them all.

Figure 4 shows the whole trade explicitly, on a stored $T = 400$ clean event, by sweeping the threshold for all three solvers of this paper. The efficiency panel shows the plateau structure just described: the 1BQF efficiency steps down at the endpoint band around $\tau \approx 0.28$, so the inherited $\tau = 0.35$ reads 75% — not because the filter lost the physics but because the cut sits

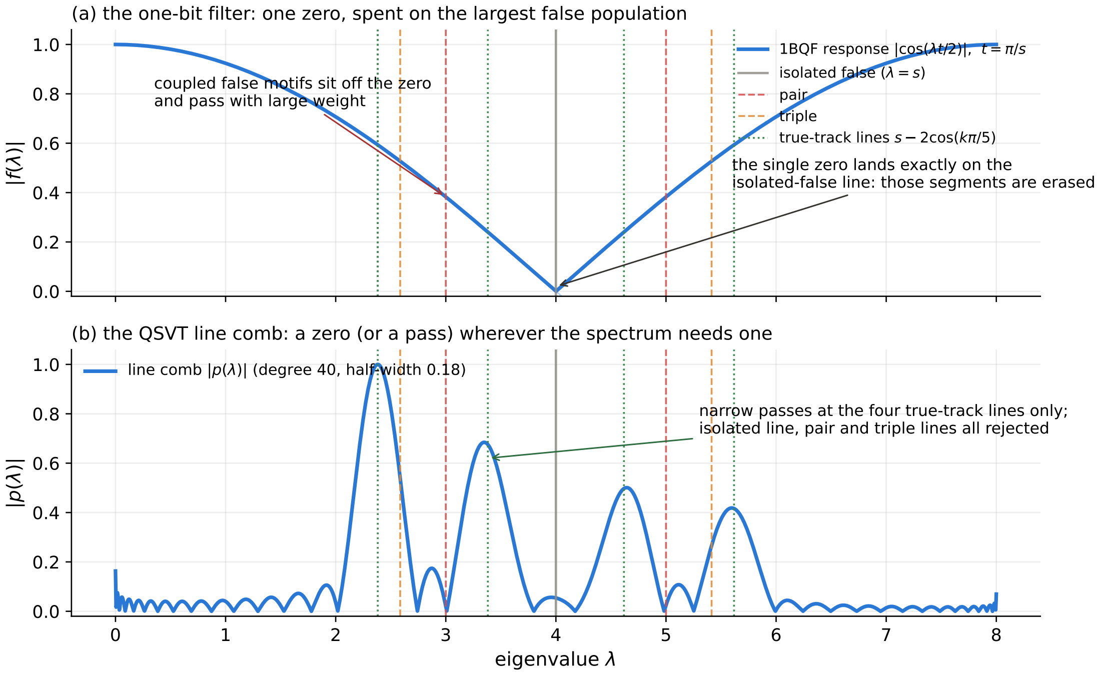


Figure 3: The two response functions against the populated spectrum ($s = 4$). (a) The 1BQF cosine: one zero, placed on the isolated-false line; the coupled false motifs (pair, triple) sit on the slopes and pass. (b) The degree-40 line comb: passes at the four true-track lines only, with the isolated, pair and triple lines all rejected.

above a reshaped true band. The false-rate panel shows what moving the threshold costs. Every solver is therefore read at its own *efficiency-first working point* (dashed verticals): the threshold just below the 1% quantile of that solver's true amplitudes, so that 99% of true segments survive by construction, with the false rate quoted as the price. This is the convention for all 1BQF headlines in this paper; the fixed $\tau = 0.35$ of eq. (8) is retained only where the classical baseline is quoted on its own established operating point. On clean fixed-acceptance events at $\gamma = 3$ the efficiency-first price is steep at high multiplicity: the 1BQF false rate climbs to several tens of percent by $T = 400$ and beyond, and section 5 quantifies this against the comb.

The second consequence is that coupled false segments pass. A false segment that accidentally couples to the graph moves off the $\lambda = s$ line, out of the zero, and into the passband. The notch has no second zero to spend on it. As density or acceptance width grows, the coupled-false population grows, and the 1BQF's false rate grows with it. The 1BQF is, precisely, the statement that unconnected fakes are exactly removable; everything in this paper beyond section 4 is about the connected ones.

On success probability: the kept-branch probability scales as $P_{\text{anc}} \approx 0.25/T$ on this problem, since the uniform input spreads over $4T^2$ segments of which order T carry surviving amplitude, so shot budgets grow linearly in T and amplitude amplification buys the usual quadratic discount.[3] These costs carry over essentially unchanged to the polynomial filters below, which is why I defer resource accounting to section 5.4.

[3]The 0.25 constant is the analytic estimate for the uniform input; the comb's analogous constant, ≈ 0.44, is directly measured (section 5.4).

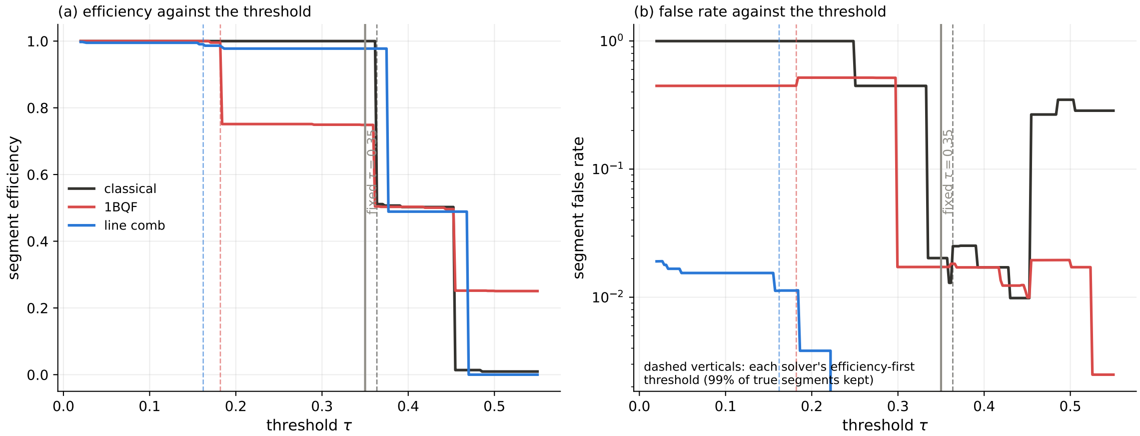


Figure 4: The threshold trade-off, on the stored $T = 400$ clean event of the canonical campaign: (a) segment efficiency and (b) segment false rate as the threshold τ sweeps, for the classical solver, the 1BQF and the line comb (quantum amplitudes on the classical signal scale). The grey vertical is the inherited fixed $\tau = 0.35$; the dashed verticals are each solver's efficiency-first working point (99% of its true segments kept). The fixed cut sits above the 1BQF's reshaped endpoint band — the origin of its 75% plateau — while the comb holds both high efficiency and a $\sim 10^{-2}$ false rate over the whole usable range.

4 From one bit to any polynomial: the QSVT construction

The promotion from eq. (11) to an arbitrary polynomial response rests on three standard ingredients: block encoding, qubitization, and a linear combination of unitaries (LCU). I assemble them here in the concrete, real-arithmetic form used by the simulations. Readers who trust the QSVT literature [7–10] can skim to the resource summary in section 4.3; the details are included because the paper's cost claims (qubit counts, walk calls, success probabilities) are read directly off this construction.

4.1 Block encoding and the qubitization walk

Rescale the Hamiltonian affinely so its spectrum fits in $[-1, 1]$: $X = (A - c_0 I)/c_1$ with c_0 the spectrum centre and c_1 slightly more than its half-width[4]. A block encoding is a unitary that hides X in its top-left corner:

$$U = \begin{pmatrix} X & \sqrt{I - X^2} \\ \sqrt{I - X^2} & -X \end{pmatrix}, \tag{12}$$

acting on the segment register plus one *dilation* qubit. The picture I find most useful is a rotation in each eigenplane: for every eigenvalue x of X there is a two-dimensional invariant subspace in which U rotates by the angle $\arccos x$.

The word *walk* deserves an explanation, because nothing in this paper has moved yet and the term is imported jargon. A walk operator is simply a fixed unitary that one applies over and over, taking one "step" per application, in analogy with a classical random walk where one transition matrix is applied repeatedly. The useful property is what repetition does. Because U rotates each eigenplane by $\arccos x$, applying the walk k times rotates it by $k \arccos x$, and the component that ends where it started is $\cos(k \arccos x)$ — which is precisely the k-th Chebyshev polynomial $T_k(x)$. One application of the walk knows only x; k applications know $T_k(x)$. Repetition of a

[4] The implementation pads the user-supplied spectral bounds by a few percent; an early campaign bug taught us that the polynomial's fit domain must exceed the padded bounds or the constructor rightly refuses.

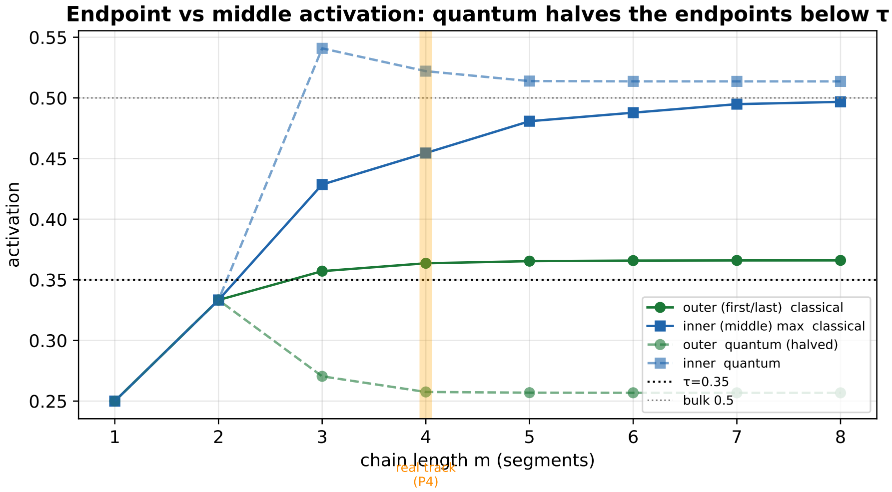


Figure 5: The chain ladder and the endpoint halving. Classical outer activation climbs eq. (7) toward 0.366 (above τ); the 1BQF response drops the P_4 outer level to 0.258, below the fixed threshold. This is the origin of the fixed-τ efficiency plateau, and the reason 1BQF headlines use the efficiency-first working point.

single fixed circuit therefore *generates a polynomial family*, one degree per step, and that is the entire mechanism by which a quantum computer applies polynomials of a matrix without ever diagonalising it. Formally, the *qubitization walk* $W = (Z \otimes I)\,U$ (the extra sign Z on the dilation qubit turns the reflections of U into clean rotations) has the defining property

$$\langle 0|\, W^k \,|0\rangle_{\mathrm{dil}} = T_k(X), \tag{13}$$

where T_k is the k-th Chebyshev polynomial and the bra-ket projects the dilation qubit onto $|0\rangle$. Equation (13) is exact, with no Trotterization and no error terms, and every operator in it is real, so the entire simulation can run in real arithmetic on orthogonal matrices. Walking k steps and looking at the undeflected component applies the k-th Chebyshev polynomial of the Hamiltonian; that is the whole theorem.

4.2 Summing the series: LCU

A degree-d response $p(X) = \sum_{k=0}^{d} c_k T_k(X)$ is then a weighted sum of walk powers. Introduce an index register of $m = \lceil \log_2(d+1) \rceil$ qubits, prepare $|\mathrm{prep}\rangle \propto \sum_k \sqrt{|c_k|}\,|k\rangle$, apply W^k controlled on the index (in practice: controlled powers W^{2^j}), undo the preparation, and post-select the index and dilation registers on zero. The surviving register state is $\propto p(X)|u\rangle$ with success probability

$$P_{\mathrm{anc}} = \frac{\big\| p(X)\,|u\rangle \big\|^2}{\big(\sum_k |c_k|\big)^2}, \tag{14}$$

the usual LCU price: the one-norm of the coefficient vector. The 1BQF is recovered, exactly, as the $d = 1$ row of this table (cos is T_1 of the appropriately rescaled operator), which is the formal content of the claim that this paper continues rather than replaces its predecessors.

Simulation-side, eq. (13) gives a matrix-free evaluator: $p(A)\mathbf{b}$ by the Chebyshev three-term recursion on sparse matrix-vector products, $\mathcal{O}(d \cdot \mathrm{nnz}(A))$ flops. In a persisted benchmark of the production configuration the $d = 40$ comb solve runs in 1.2–1.7 s at $T = 1000$ ($n = 4 \times 10^6$), and the evaluator agrees with the full qiskit circuit and with a gate-streaming evaluation to better than 10^{-14} at circuit-affordable size (three-way maximum discrepancy 8×10^{-15}). All large-T results below use this exact evaluator; nothing in this paper relies on an approximate simulation of the quantum readout.

4.3 Resource summary

For a degree-d filter on an event of T tracks:

- **Width**: $n_{\text{sys}} = \lceil \log_2 4T^2 \rceil$ system qubits $+1$ dilation $+m = \lceil \log_2(d+1) \rceil$ index qubits. With the production degree $d = 40$ this is 21 qubits at $T = 50$ rising to 29 at $T = 1000$. Width is cheap, and grows only logarithmically.
- **Depth**: d controlled walk calls per coherent repetition, times $\mathcal{O}(1/\sqrt{P_{\text{anc}}})$ amplitude-amplification rounds.
- **Success**: $P_{\text{anc}} \propto 1/T$ (measured constant ≈ 0.44 for the comb of section 5), so total walk calls scale as $d\sqrt{T}$.

The quantum content of this paper is this resource profile; the response functions themselves are classically simulable by the same recursion, and I return to what that does and does not concede in section 10.5.

5 Fitting the polynomial to the clean events: the line comb

5.1 The design insight: the true spectrum is discrete

Here is the observation that turns QSVT from a mathematical possibility into a design tool. On this detector geometry every true track has exactly five hits, hence exactly four segments, hence, by eq. (9), exactly the same four eigenvalues. The true spectrum of the entire event, however many thousand tracks it contains, is four discrete lines. We are not trying to pass a region of the spectrum; we are trying to pass four known radio stations, and everything between them is static.

My first design did not understand this. A band-limited approximate inverse ($\approx 1/\lambda$ across a contiguous band containing the true lines, zero outside) works beautifully at low multiplicity and then fails exactly where it matters: as T grows, accidental tangles of false segments produce eigenvalues throughout the band, and a band design passes them all. At fixed threshold its false rate reached tens of percent by $T = 400$, worse than classical.[5] The production design is therefore a *line comb*: narrow Gaussian passes of half-width h_w, weighted $\propto 1/\lambda$, centred at the four lines $s - 2\cos(k\pi/5)$ only, fitted as a bounded Chebyshev series of degree d. The defaults, fixed once by a two-dimensional (d, h_w) scan and never tuned per event, are $d = 40$, $h_w = 0.18$.

Why does a width of 0.18 work? The nearest systematically-populated false lines, the P_3 bridge modes at $\{s - \sqrt{2},\ s,\ s + \sqrt{2}\}$, sit $\gtrsim 0.2$ away from the true lines, so the comb threads the gap; and the resolution law of a degree-d Chebyshev series, features no narrower than $\approx 2W/d$ on a domain of half-width W, makes $d = 40$ sufficient for $h_w = 0.18$ on the clean-event domain. Degree, width, and line spacing are one budget, and this is the budget the noisy events of section 9 will re-open.

5.2 Clean-event results

On the fixed-acceptance ($\varepsilon = 2\,\text{mrad}$), $\gamma = 3$ study, the canonical benchmark of the programme, with events and solutions shared segment-for-segment across all three solvers through a common data pipeline, the picture at the fixed threshold $\tau = 0.35$ is:

- **Classical**: efficiency 100% throughout (the minimum true amplitude $4/11$ clears τ at every T); false rate grows from ≈ 0 at low T to 2.25% at $T = 400$ and 20.5% at $T = 1000$ as the $\mathcal{O}(T^2)$ false pairing feeds the active set.

[5] A useful failure: it taught us that the exploitable structure is the discreteness of the geometry-pinned true spectrum, not its location.

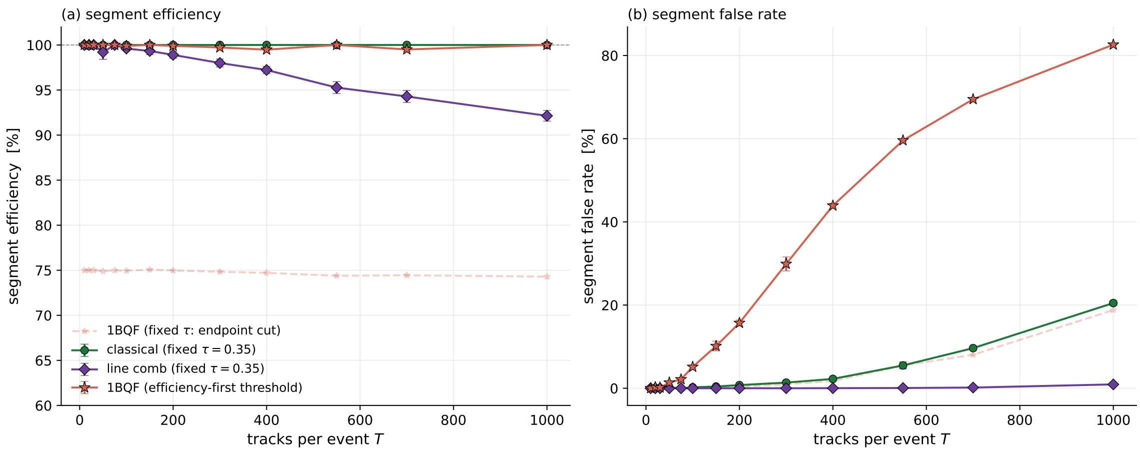


Figure 6: The clean-toy headline ($\gamma = 3$, fixed $\varepsilon = 2\,\mathrm{mrad}$, no hit drop): (a) segment efficiency and (b) segment false rate against multiplicity, for the classical solver and the line comb at the fixed threshold and the 1BQF at its efficiency-first working point (its fixed-τ curve is kept faded: the $\approx 75\%$ plateau is the endpoint cut of section 3, not lost physics). Error bars are standard errors over event repetitions.

- **1BQF**: the fixed-τ efficiency plateau of section 3 (74.7% at $T = 400$, the endpoint cut); false rate at or just below classical (1.73% at $T = 400$, 18.8% at $T = 1000$).
- **Comb**: efficiency 97–100% with false rate $\leq 0.02\%$ through $T = 400$ (zero to a few false actives per event where the classical solver admits tens), rising to 0.17% at $T = 700$ and 0.9% at $T = 1000$ against the classical 20.5%, at efficiency 92.1%.

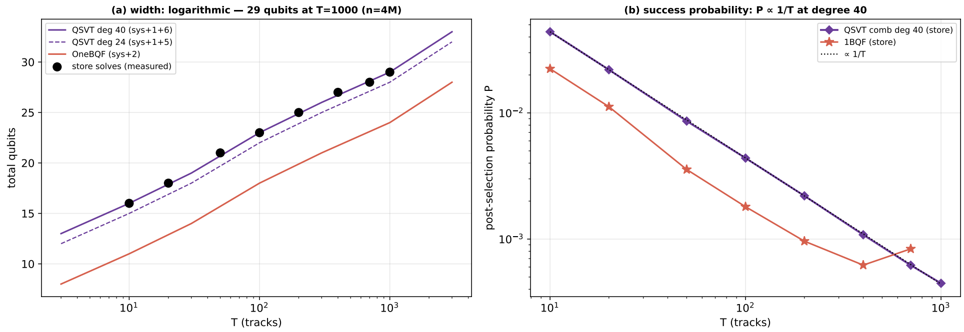
At the efficiency-first working point the comb wins clearly through $T = 400$ (1.3% vs classical 3.1% on matched fresh solves), but at extreme density the composition effect dominates everyone: the false-to-true ratio $n_F/n_T \approx T - 1$ converts even excellent per-segment separation into a growing impurity, and by $T = 1000$ the efficiency-first false rates cross ($\approx 21\%$ comb vs 17.5% classical, crossover near $T = 700$).[6] The two regimes, fixed-τ where the comb wins by an order of magnitude and more, and the strict efficiency-first point at extreme density where composition flattens everyone, bracket the operational truth.

The comb's amplitude picture explains the numbers. Under the comb response the false population collapses to $\lesssim 10^{-2}$ of the true median (the passes reject everything off the lines), while the true band stays tight around its classical profile: the $1/\lambda$ weighting inside the passes preserves the chain shape rather than reshaping it as the single cosine does. Where the 1BQF splits the true population and the classical solver drags a false tail across the threshold, the comb simply leaves a gap.

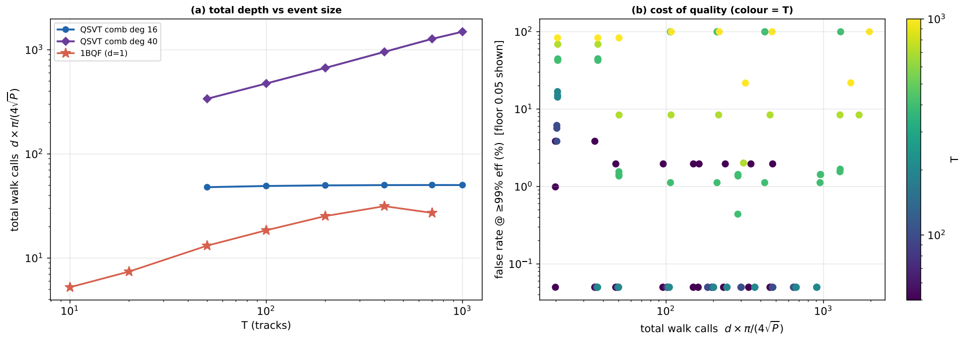
5.3 Hit inefficiency: the first appearance of fragments

Dropping 1% of hits breaks $\approx 5\%$ of tracks into fragments: chains of one, two, or three segments whose spectra (section 2.5) are *not* on the four P_4 lines. The comb, by design, does not pass them. Per-class efficiency under the production comb is 100% on intact tracks and 0% on every fragment class, which is why its overall efficiency reads 94.0% rather than 97–98% at $T = 400$ under drop. Extending the comb with passes at the fragment lines does not help: it re-admits their false spectral twins wholesale (a P_2 fragment and a false pair are the same matrix, figure 8), so fragments are a spectral floor *on this Hamiltonian*, the first appearance of the twin obstruction

[6] The crossover is robust; its size is aggregation-dependent. A whole-store aggregate reads the classical efficiency-first false rate at $T = 1000$ as low as 10%. Both sources agree the comb no longer wins at the efficiency-first point beyond $T \approx 700$, which is the operative statement.



(a) Register width and success probability.



(b) Total walk calls (depth).

Figure 7: Measured resource scaling of the comb on the shared events: width grows logarithmically (21 qubits at $T = 50$ to 29 at $T = 1000$); success probability falls as $\approx 0.44/T$; amplified walk calls grow as $d\sqrt{T}$, with the half-resolved $d = 16$ comb holding a T-independent $\approx$ 50-call budget to $T = 700$.

that section 9 meets at full strength — and that section 9.4 shows an operator modification can lift. Recovery on the unmodified Hamiltonian must come from information a spectral filter cannot see, namely hit-level constraint information, and that is classical post-processing, outside the scope of this paper.[7]

5.4 Resource scaling and the depth ladder

Measured on the shared events: total width 21 qubits at $T = 50$ to 29 at $T = 1000$; success probability $P_{\text{anc}} \approx 0.44/T$; walk calls per amplified repetition $d \cdot \pi/(4\sqrt{P_{\text{anc}}}) \propto d\sqrt{T}$. Two findings sharpen the budget. A half-resolved comb at $d = 16$ matches the $d = 40$ metrics up to $T = 700$ at an approximately T-independent $P_{\text{anc}} \approx 6\%$, about fifty total walk calls, flat in T, because partially-merged passes still cover the four lines while rejecting the bulk. And the degree axis is not monotonic: intermediate degrees ($d \approx 20$–32) can cliff at the efficiency-first working point through Chebyshev fit pathologies, so every production degree must be validated, not interpolated.

5.5 Spectral twins and the floor theorem

The most important structural limit on the whole filter family deserves its own name and its own picture. Call two segment clusters *spectral twins* when their compatibility graphs are isomorphic: the same number of segments, connected in the same pattern, regardless of where the segments

[7] An earlier version of this study paired a hit-occupancy recovery gate with the comb alone; it was retired as mission creep (no other solver received the same help, and the tool is orthogonal to the spectral question). Classical post-selection is developed and reported separately in the programme's occupancy study.

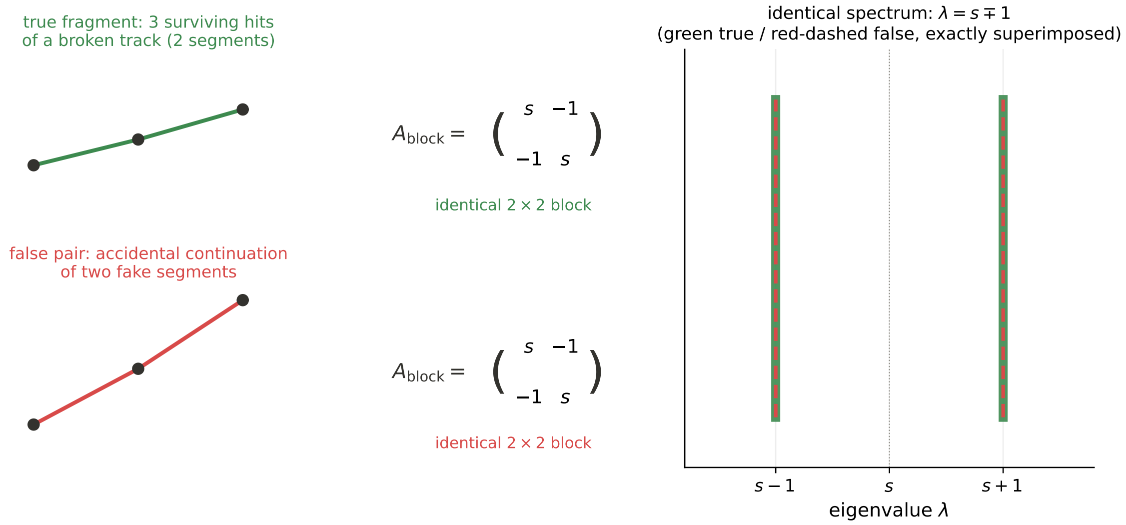


Figure 8: Spectral twins. A true fragment of a broken track (top) and an accidental false pair (bottom) have isomorphic compatibility graphs, hence identical 2×2 sub-blocks of A, hence exactly the same eigenvalues and amplitudes. Every function of A treats them identically; this is the floor theorem, and under noise it becomes the fragment floor of section 9.

physically are or whether they belong to a real track. Because A acts block-diagonally over connected components, each cluster contributes a sub-block of A, and twin clusters contribute *the same sub-block up to relabelling* (permutation similarity). The consequence is the *floor theorem*: for every function f, $f(A)\mathbf{b}$ assigns twin clusters identical amplitudes, because permutation similarity commutes with matrix functions and $\mathbf{b}$ is uniform on every cluster. No response, fitted or otherwise, and no classical amplitude scoring, can tell twins apart: whatever passes one passes the other.

Figure 8 draws the pair of twins that will matter most in this paper. A true two-segment fragment (three surviving hits of a broken track) and a false pair (two fake segments that happen to continue one another inside the acceptance) are physically completely different objects; but both are two segments sharing one hit, both produce the same 2×2 sub-block of A, and both therefore sit on exactly the eigenvalue lines $s \mp 1$ with exactly the same amplitudes. On clean events twins are rare (an exhaustive scan of the $T \leq 200$ events finds none; 0.6% of true segments by $T = 400$ with the acceptance widened to 4 mrad), but they are the true ceiling of the design, and under noise they return, as a population, in section 9.

Two further structural results complete the clean-toy picture. The *shared-line theorem*: the odd modes of a contaminated five-segment chain coincide exactly with the $s \pm \sqrt{3}$ lines of the three-pronged hub $K_{1,3}$, so no comb can null both populations independently; pass menus trade one against the other with a fixed exchange rate. And the *mirror tax*: a chain-reflection symmetry forces the false extension of a recovered cluster to the recovered-true amplitude level, so recovery passes always buy their efficiency at a false-rate premium. Together these results closed the clean-toy design question. The production comb is optimal up to provable exceptions, and further progress must come from changing the operator or leaving the spectral domain, which is exactly where the rest of this paper goes.

6 The heavy-noise operating point

From here on the operating point is the heavy configuration: $\sigma_{\text{scatt}} = 10^{-4}$, $\sigma_{\text{res}} = 20\,\mu$m, hit drop 1%, at $T = 200$, with the acceptance set by the kink formula to $\varepsilon = 6.3$ mrad. Three reps of every event are used throughout; where a single event is dissected, it is rep 0.

The acceptance width is the story. At $\varepsilon = 2$ mrad a false segment rarely finds an angular partner; at 6.3 mrad the compatibility graph fills with small false structures. Two facts about that graph, both established by direct decomposition of the solved events, control everything downstream:

1. **There is no giant tangle.** The compatibility graph at heavy $T = 200$ splits into 869 nontrivial connected components of at most 17 segments, floating in a sea of 155,371 isolated segments (rep 0). Noise, at this operating point, buys many small motifs rather than one big mess, and small components mean the entire spectrum is computable exactly, component by component, with dense eigensolvers. Every spectral statement in sections 7–9 is exact, not approximate. Figure 9 shows what these components actually look like in the detector: the component-size distribution, and real examples drawn from the event, from a pure-false pair to the largest seventeen-segment tangle. Seen at the micron scale of their internal kinks, the "tangles" are bundles of near-collinear competitors — exactly the structures the widened acceptance admits.
2. **The false actives are motif-borne.** At the classical efficiency-first working point the event has 2238 active segments of which 1454 are false; those false actives live in 707 components that contain, between them, only 142 true segments. Ninety percent of the false-active population sits in essentially pure-false motifs (pairs, triples, short bridges), not entangled with true tracks at all. Spectrally, this population lands exactly where the motif dictionary of figure 2 says it must: figure 10 shows the measured per-mode solution weight of the event against λ, with the false weight piled up on the pair line $s - 1$ and the triple lines $s \pm \sqrt{2}$, interleaving but not touching the four true lines.

The solver scoreboard at this point makes uncomfortable reading. At the efficiency-first working point the classical false rate is 0.65; the 1BQF's is 0.65 as well, since the notch's one zero still annihilates the isolated bulk while every false *active* is by definition coupled, off the line, and untouched. The clean-event comb, designed for a 2 mrad world where false structures barely exist, fails too: the wide acceptance populates false motif lines (figure 10) inside its pass windows. The important statement here is the diagnosis: at heavy noise all three solvers fail for the same reason, a false population that has moved onto spectral lines the designs were never told about — and the cure, developed in section 9, is to measure those lines and refit the response to them. The classical literature also offers two Hamiltonian terms as candidate cures, occupancy and bifurcation. The next two sections take them in turn; neither behaves the way its classical reputation suggests, and neither can be judged separately from the refit.

7 The occupancy term

7.1 The term

Denby's original network [5] constrains each hit to be used by at most one selected segment per role. The quadratic form of that constraint, per-hit occupancy, adds $E_{\text{occ}} = \alpha \sum_h (o_h - 1)^2$ with $o_h = \sum_{s \ni h} x_s$, which at the matrix level reads

$$A_{\text{occ}} = A + 4\alpha I + 2\alpha B_{\text{all}}, \qquad \mathbf{b} \to \mathbf{b} + 4\alpha \mathbf{1}, \tag{15}$$

where B_{all} couples every pair of segments sharing a hit in the same role: two segments departing the same hit, or two arriving at it. Consecutive segments of one track share their interior hit in

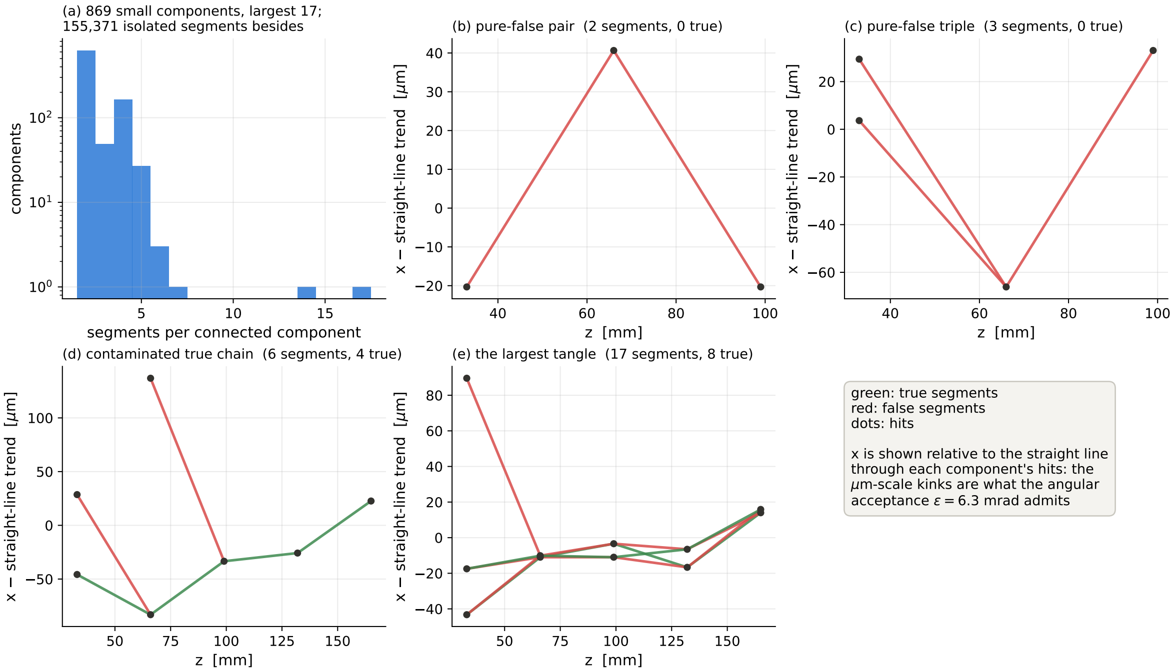


Figure 9: The heavy-noise compatibility graph, seen directly (heavy $T = 200$, rep 0). (a) Component-size distribution: many small motifs, no giant component. (b–e) Real components drawn in the detector, with the straight-line trend subtracted so the micron-scale kinks are visible: a pure-false pair, a pure-false triple, a true chain with false contamination, and the largest tangle in the event (17 segments, 8 true). Green segments are true, red false.

opposite roles and are not coupled. The isolated attractor moves to $(\delta + 4\alpha)/(\gamma + \delta + 4\alpha)$ and the threshold tracks it.

7.2 Classically it works; the notch it destroys

On the heavy $T = 200$ benchmark event the in-matrix occupancy term at $\alpha = 0.3$ does exactly what Denby promised: the classical efficiency-first false rate falls from 0.650 to 0.210 at efficiency 0.991. If this paper were classical, the section would end here with a recommendation.

For the 1BQF it is a disaster, and the mechanism is worth spelling out because it generalises. The notch's power is that isolated false segments sit exactly on its zero. The occupancy coupling $2\alpha B_{\text{all}}$ connects precisely those segments to their co-hit neighbours, which is its job, and thereby moves them off the zero. On the benchmark event the co-hit false bulk, notched to amplitudes $\sim 10^{-19}$ under the base Hamiltonian (99% of it below 10^{-17}), re-emerges under A_{occ} at amplitudes on the true band: its 99th percentile rises to 1.5×10^{-3} against a true median of 1.2×10^{-3}. The rank separation collapses (1BQF AUC 0.996 $\to$ 0.634) and no threshold rescues it; the full sweep shows every occupancy-1BQF operating point dominated by the base-1BQF curve. Nor does the self-coupling γ offer an escape: the notch tracks the diagonal, distance-to-notch is γ-invariant, and at the window-restoring $\gamma = 236$ the occupancy system is strictly worse. Post-selection tells the same story: the occupancy term diverts the ancilla-conditioned probability mass almost entirely onto false segments.

The mechanism generalises to any *fixed* response: a spectral filter earns its living from populations pinned to known eigenvalues, and any Hamiltonian term that couples a pinned population unpins it. For the 1BQF the verdict really is final, because the notch has no further freedom to spend: its response family is exhausted by the single parameter t, the retune $t = \pi/s'$ is already included above, and figure 11(c) shows the γ direction offers no escape either. As a Hamiltonian term,

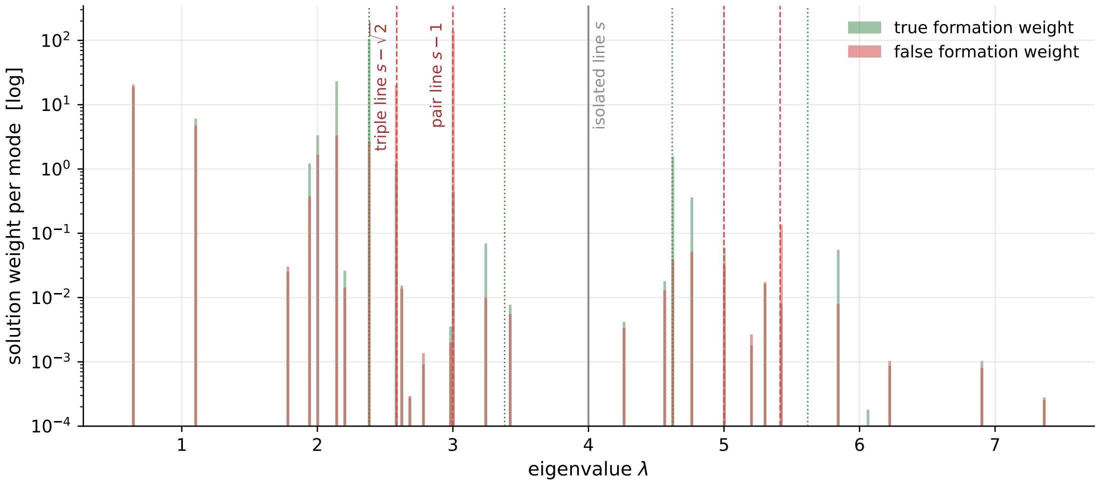


Figure 10: The measured spectral pile-up of the heavy $T = 200$ event (rep 0), computed exactly by per-component diagonalisation: solution weight per mode against λ, split into true (green) and false (red) content. The false weight concentrates on the pair line $s-1$ and the triple lines $s \pm \sqrt{2}$ of the motif dictionary (figure 2), close to but never on the four true-track lines (green dotted).

occupancy is poison to a notch, and the classical intuition for the term does not transfer to it.[8] But "final for the notch" is not "final for the filter family". The fork term in the next section will teach the general lesson of this paper — that an operator modification can only be judged together with a response refitted to the spectrum it creates — and section 9.4 returns to the occupancy term with exactly that machinery. The verdict there reverses: what destroys the degree-one filter turns out to be the single most helpful operator modification in the study once the polynomial is allowed to follow it.

8 The fork term and the moving spectrum

8.1 The term and its validity envelope

The second Denby–Peterson penalty targets *bifurcations*: two segments claiming the same hit in the same role at small mutual angle, the fork topology that produces the closely-competing candidates a wide acceptance admits. The ε-windowed form adds

$$A' = A + \beta\, B_{\text{fork}}(\varepsilon_B), \tag{16}$$

where B_{fork} couples co-hit same-role pairs within angular window ε_B (default: the acceptance itself). Unlike the occupancy term, the fork is surgical where it matters. On clean events it touches $\approx 2\%$ of segments, and provably no pure-true chain carries a fork edge, so the true comb lines are β-invariant. Earlier classical-side work in this programme established that the fork migrates contaminated-true spectral weight back toward the comb bands; whether real solvers can collect that migration is settled in section 9.

Adding a positive semi-definite coupling widens the spectrum, and the solver imposes two validity conditions with closed forms. Positive definiteness requires $\gamma > \gamma_{\text{pd}} = \gamma_{\text{ref}} - \lambda_{\min}$; the one-period condition of any cosine-family filter (the top of the spectrum must not alias through the response period) requires $\gamma > \gamma_{\text{win}} = \lambda_{\max} - \gamma_{\text{ref}} - 2\delta$, both extremes measured at a reference γ_{ref} and

[8] The occupancy constraint can also be applied entirely outside the Hamiltonian, as a classical hit-uniqueness post-selection applied evenly to every solver's output; the programme develops that tool in a separate study, deliberately kept out of this paper, whose subject is the spectral side.

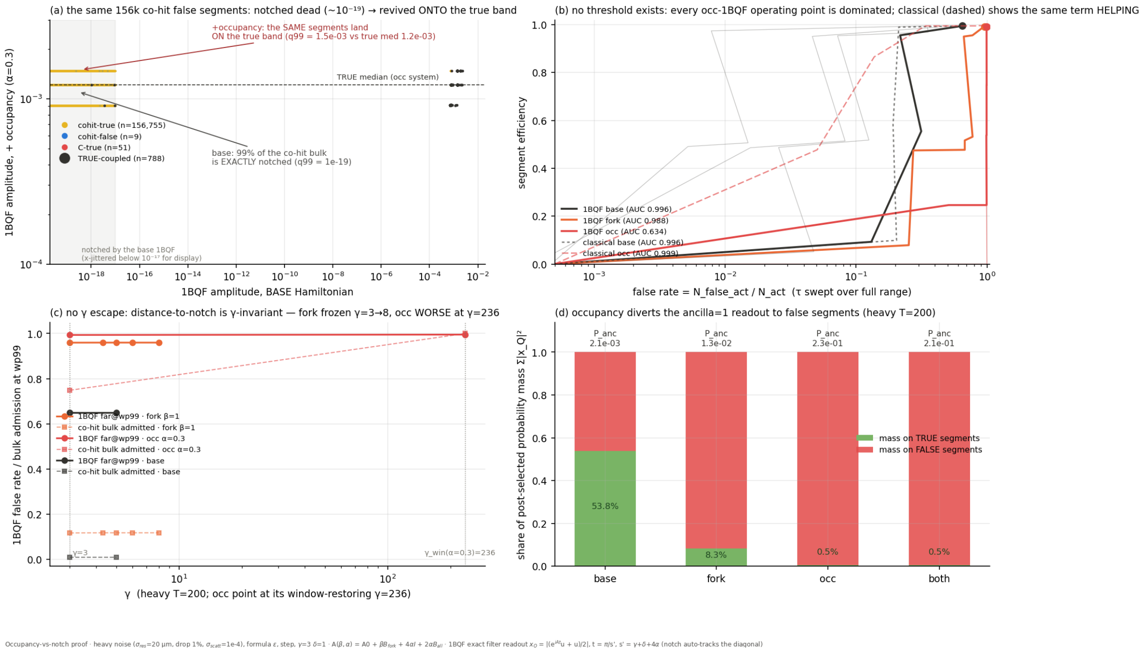


Figure 11: The occupancy proof (heavy $T = 200$): (a) the same co-hit false segments, exactly notched under the base Hamiltonian, re-emerge on the true amplitude band under A_{occ}; (b) no threshold escape: every occupancy-1BQF operating point is dominated, while the same term helps the classical solver; (c) no γ escape; (d) the post-selected probability mass diverts to false segments.

shifted rigidly, since γ enters only the diagonal. At heavy $T = 200$ the measured requirement is $\gamma^* \approx 3.4$ at $\beta = 0.5$ and ≈ 4.4 at $\beta = 1$. The notch tracks the diagonal automatically ($t = \pi/s'$, $s' = \gamma + \delta$), and the comb is rebuilt on the measured shifted spectrum. Every fork result below carries these retunes; a window-violating control ($\beta = 0.5$ at $\gamma = 3$) is kept in the study to show they matter.

8.2 The operator and the response are one design

One methodological point, learnt the hard way, belongs here rather than in a results table. Retuning a *fixed* response to the fork system — tracking the diagonal, rebuilding the comb on the shifted domain — is not enough, because the retunes move the response's centre and domain but never ask where the false weight now lives. Evaluated that way, both Denby–Peterson terms read as regressions, and the reading is a statement about the mismatch between an old response and a new spectrum, not about the terms. The correct procedure, used for every operator result in this paper, is to measure the modified system's false pile-up and refit the polynomial to it: change the Hamiltonian, and the activations and their spectral support move; when they move, the polynomial must be refitted to the moved spectrum. Carrying out that refit, for the fork and then for the occupancy term, is the subject of section 9. (A second, smaller point: at heavy noise the efficiency-first convention degenerates for an unrefitted comb, whose contaminated-true amplitudes collapse toward zero, so sections 9 onward compare solvers at matched efficiency instead.)

8.3 A Trotter aside that will matter on hardware

Everything in this paper so far treats the evolution e^{iAt} as exact. On hardware it is not: a quantum computer realises e^{iAt} by a circuit, and the standard cheap construction, the *product*

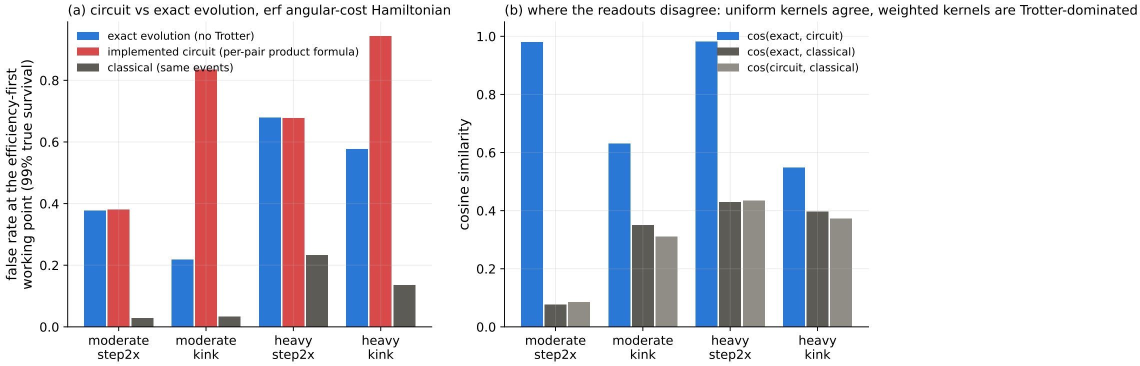


Figure 12: The exact-evolution arbiter on the erf angular-cost Hamiltonian (heavy and moderate noise, $T = 200$). (a) False rate at the efficiency-first working point: exact evolution against the implemented product-formula circuit and the classical baseline on the same events. Uniform-weight kernels (step2x) show circuit = exact, so the remaining deficit is physics; weighted kernels (kink) are Trotter-dominated. (b) The same story as readout fidelity: cosine similarity between the three solutions.

formula (or Trotterization), builds it from the pairwise pieces of the Hamiltonian. The idea is simple: A is a sum of many small commuting-or-not terms, one per coupled segment pair, and while $e^{i(X+Y)t}$ is hard, the product $e^{iXt}e^{iYt}$ is easy — one small gate per term. If X and Y commute the product is exact; if not, it errs at order $t^2[X, Y]$, and one hopes the accumulated error stays below the physics. The 1BQF circuit of ref. [3] is exactly such a per-pair product.

The programme's measurements show where that hope holds and where it breaks, and the split is clean (figure 12). With *uniform* coupling weights (every term the same strength, as in the step-kernel Hamiltonian of this paper) circuit and exact evolution agree to cosine similarity ≈ 0.985: the product formula is safe. With *heterogeneous* weights (the erf-weighted angular kernel, or a fractional fork $\beta = 0.5$ riding on unit compatibility edges) the per-pair product scatters the readout badly enough to dominate the physics: on the fork system the circuit reads false rate 0.96 where the exact evolution reads 0.73, and on the erf kernel removing the product formula alone collapses the false rate from 0.83 to 0.22 at moderate noise. The rule of thumb for hardware is that heterogeneous coupling weights break the naive product formula, and circuit realisations of weighted Hamiltonians need the structured synthesis route (edge-coloured pair ordering) developed in the programme's direct-structural-synthesis study. All fork and fit results in this paper use exact evolution precisely so that filter physics and circuit error are never conflated; section 10.3 returns to what this means for deployment.

9 Fitting the polynomial to noisy events: roots at the measured pile-up

9.1 Measuring the false pile-up

The fit begins with a measurement no fixed design ever asked for: where, exactly, does the false amplitude come from? Because the heavy-noise compatibility graph has no giant component (section 6), the per-component eigendecomposition gives, for every eigenvalue in the event, its exact content: how much of the mode's weight lies on true segments, how much on false, and how much of each segment's amplitude it builds. Under any response p,

$$x_s(p) = \sum_i p(\lambda_i)\,(\mathbf{v}_i^\top \mathbf{b})\,v_i(s), \tag{17}$$

linear in p, so the question of where the false amplitude comes from is well posed and exactly answerable. The answer, on the heavy $T = 200$ events, is concentrated to a degree I did not expect. The false-active formation weight piles up on the *pair line* $\lambda = s - 1$: on the three events measured, that single bin carries 1122–1378 weight units against 4–10 units of true weight, with the triple lines $s \pm \sqrt{2}$ next; the overall bin-level overlap between false-active and true formation is 0.014–0.023 in cosine. The false population is not smeared. It is parked on the eigenvalue lines of identifiable two- and three-segment motifs, almost none of which are shared with true formation.

9.2 Fitting the response

With eq. (17) linear in the response, the fit is a regularised least-squares problem over binned response values ($\Delta\lambda = 0.02$):

$$\min_{p} \sum_{s \in \text{false}} x_s(p)^2 \;+\; \mu \sum_{s \in \text{true}} \left(x_s(p) - x_s^{\text{cls}}\right)^2, \qquad |p| \le 1, \tag{18}$$

with the isolated-false line pressured to zero and μ swept to trace the efficiency–purity trade. No root position is chosen by hand: zeros appear where false weight concentrates because that is what the objective rewards. The fitted response reproduces the comb's passes at the (contamination-shifted) true lines and drives deep roots through $s - 1$ and $s \pm \sqrt{2}$: precisely the polynomial that no clean-event design could have anticipated, because the root locations are properties of the measured noise.

9.3 Results

Table 1 is the main result of the section, and it is worth being explicit about how to read it, because every number in it is the same kind of object. Fix a system (a row block: the base Hamiltonian, or the fork system at its validity point) and a target efficiency (the "eff" column). For each solver, slide that solver's own threshold until exactly the target fraction of true segments survives — the matched-efficiency convention of section 2.4 — and record the false rate of the selected sample at that threshold. Lower is better; solvers in the same row are compared at identical efficiency by construction, so the false rate is the *entire* difference between them. All numbers are means over three events, heavy noise, $T = 200$.

system	eff	classical	1BQF	**fitted response**
base ($\beta = 0$)	0.98	0.124	0.199	**0.096**
base ($\beta = 0$)	0.97	0.124	0.199	**0.061**
fork ($\beta = 0.5, \gamma^*$)	0.98	0.463	0.145	**0.085**
fork ($\beta = 0.5, \gamma^*$)	0.97	0.292	0.124	**0.046**
either	0.99	0.63–0.76	0.65–0.96	0.63–0.77

Table 1: Segment false rate at matched efficiency (lower is better): each solver's threshold is set so the stated fraction of true segments survives, and the false rate of the resulting selection is quoted. Heavy noise, $T = 200$, three-event means. "Fitted response" is the optimal binned response of eq. (18), refitted to each system. The last row is the fragment floor of section 9.3: at $\varepsilon_{\text{seg}} = 0.99$ every solver floods, on the base and fork systems alike.

Three conclusions follow.

First, the expected ordering is restored. Below efficiency ≈ 0.985 the fitted quantum filter sits under the classical baseline everywhere: 0.096 against 0.124 at $\varepsilon_{\text{seg}} = 0.98$, 0.061 against 0.124 at

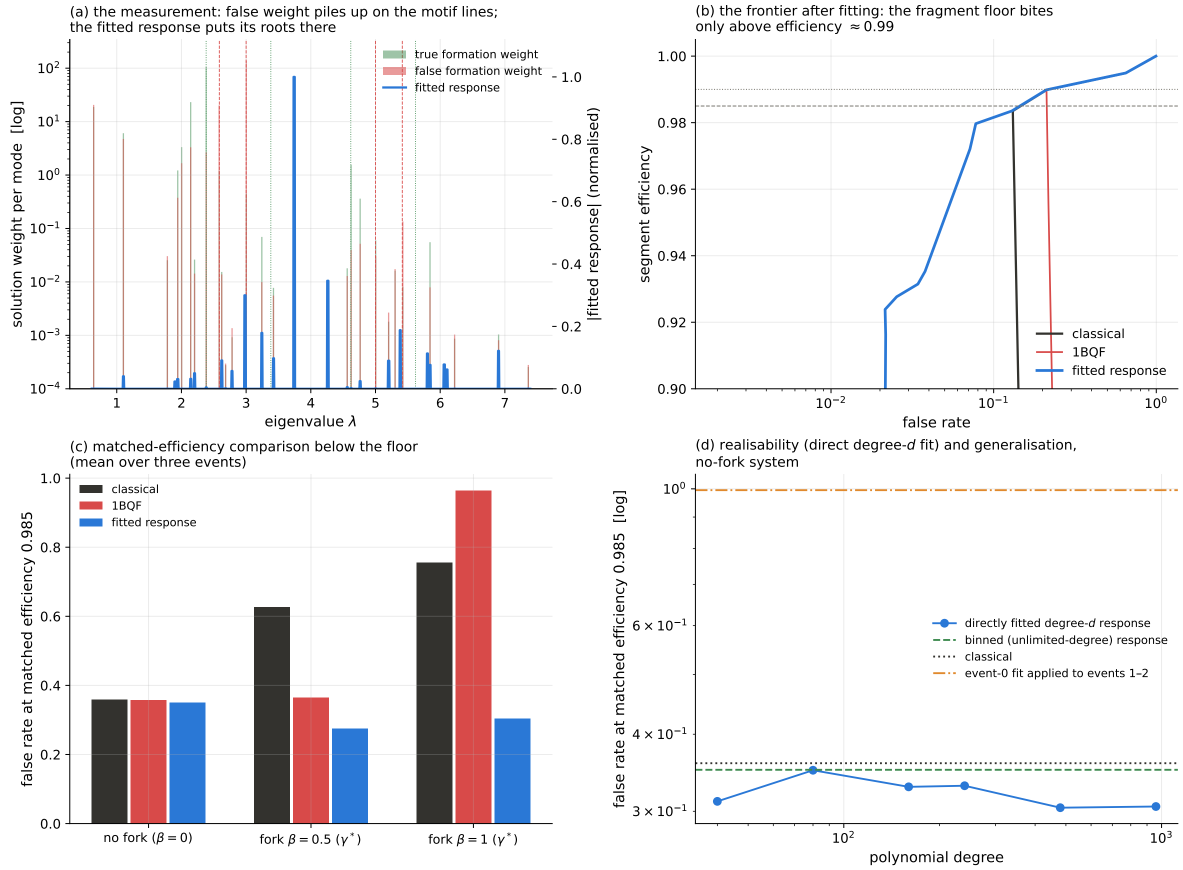


Figure 13: Fitting the response to the measured pile-up (heavy $T = 200$): (a) exact per-mode false (red) and true (green) formation weight against λ, with the fitted response overlaid; the fitted roots land on the pair and triple lines. (b) The resulting efficiency/false-rate frontier, with the fragment floor biting only above $\varepsilon_{\text{seg}} \approx 0.99$. (c) False rate at matched efficiency 0.985 for the base and fork systems (three-event means). (d) The same quantity for responses fitted directly at fixed polynomial degree: production-scale degrees already realise the measured optimum (section 10.1).

$\varepsilon_{\text{seg}} = 0.97$. The claim this section set out to test, that the false rate should still be lower for the quantum filters once they are fitted, holds.

Second, the fork earns its keep, but only through the refit. On the fork system the fitted response reaches 0.046 at $\varepsilon_{\text{seg}} = 0.97$, beating the fitted no-fork system's 0.061, while the classical solver on the same fork system reads 0.292. The operator knob does exactly what the spectral-migration analysis promised, provided the response follows it: changing the Hamiltonian without refitting the polynomial is not a half-measure, it is worse than doing neither (section 8.2). The standard-notch 1BQF also benefits from the fork below the floor (0.145 against 0.199 at $\varepsilon_{\text{seg}} = 0.98$): the fork pushes coupled-false modes away from the passband even when the only zero sits at s'.

Third, the wall at $\varepsilon_{\text{seg}} \geq 0.99$ is real on these systems, and now has a name. At the strict 99% point everything, classical included, floods back to $f_{\text{seg}} \approx 0.65$. The per-component decomposition locates the cause exactly: about 2% of *true* segments — a couple of dozen per event, counted directly and consistently on all three events — are spectral twins of the false motifs, in exactly the sense of section 5.5 and figure 8. They are the hit-drop fragments of section 5.3: a broken track's two-segment piece *is* a pair, sitting on the pair line, in a pure-true component, plus a handful of contaminated mixed chains. Any response with a root at $s - 1$ kills them alongside the false pairs; their population exceeds the 1% efficiency budget; and once the working point must dip below their suppressed amplitudes, the entire suppressed false band comes back above threshold. This is the *fragment floor*: the noisy-event generalisation of the clean floor theorem, no longer a curiosity but the binding constraint. The robust statement is the measured wall itself: on the base and fork systems every solver holds $\varepsilon_{\mathrm{seg}} = 0.98$ and none holds $\varepsilon_{\mathrm{seg}} = 0.99$, on all three events. On *those* Hamiltonians no function of λ can cross it — but the floor is a twin degeneracy of the operator, not of the event, and the next subsection shows an operator that removes it.

9.4 The occupancy term, refitted

Section 7 ended with a debt: the occupancy verdict was passed on the notch, a filter with no freedom to follow the spectrum the term creates, and the fork has since taught us that operator terms can only be judged with a refitted response. The occupancy system makes the refit technically harder — the co-hit coupling $2\alpha B_{\mathrm{all}}$ merges the compatibility graph into a giant component, so the exact per-component measurement behind eq. (18) is unavailable — but not impossible: a degree-d response needs no eigendecomposition, because the moment vectors $T_k(X)\mathbf{b}$ are computable by sparse matrix–vector recursion and the amplitudes are linear in the Chebyshev coefficients. The fit of eq. (18) then becomes an exact least-squares problem in $d{+}1$ coefficients, over *every* segment of the event including the isolated bulk. Run identically on the base and occupancy systems (same machinery, same events, only the Hamiltonian differs), it settles the question.

The verdict of section 7 reverses (figure 14). At matched efficiency 0.98 the directly fitted response reads 0.085 on the base system and 0.050 on the occupancy system at $\alpha = 0.05$; at 0.97, 0.071 against 0.039. The term that destroys the notch roughly halves the false rate of the fitted polynomial, while the classical solver on the same systems sits at 0.12 throughout. Two structural findings come with the headline. First, *small coupling wins*: $\alpha = 0.05$ beats the classically-motivated $\alpha = 0.3$ (0.050 vs 0.067 at matched efficiency 0.98), because the occupancy term stretches the spectral span ($6.7 \to 43 \to 242$ for $\alpha = 0 \to 0.05 \to 0.3$) and a fixed degree budget loses resolution proportionally; the spectral benefit saturates long before the classical amplitude benefit does. Second, and most strikingly, *the occupancy coupling breaks the fragment floor*. One of the three events is floor-bound on the base system — its fitted response cannot do better than 0.654 at matched efficiency 0.985, because its fragment twins are exactly degenerate with false pairs — and on the occupancy system the same event fits to 0.062. The mechanism is the floor theorem read in reverse: the twin argument requires isomorphic components, and the co-hit coupling embeds every motif in its hit-sharing neighbourhood, which is different for a fragment of a real track than for an accidental false pair. The embedding breaks the isomorphism, the degeneracy lifts, and a response fitted to the new spectrum separates what was provably inseparable on the base Hamiltonian. The fragment floor of section 9.3 is therefore a property of the *base* Hamiltonian, not of the problem.

Honesty requires the same caveats as the rest of this section: these are per-event fits, the measured ceiling of a universal design rather than a deployed algorithm (section 10.2), and the degree budget that collects the occupancy win is larger than the base system's (the span costs resolution; most of the gain arrives by degree 80–160 at $\alpha = 0.05$). Whether an ensemble-fitted universal response keeps the floor-breaking property is, with the universal response itself, the programme's clear next measurement.

9.5 Why the 1BQF specifically cannot follow

A degree-one filter owns one zero, and the measurement says the heavy false spectrum has two mandatory root locations: the isolated line at s ($\mathcal{O}(10^5)$ segments) and the pair line at $s-1$ (the active pile-up). The study's control makes the conflict vivid. Retuning the notch onto the

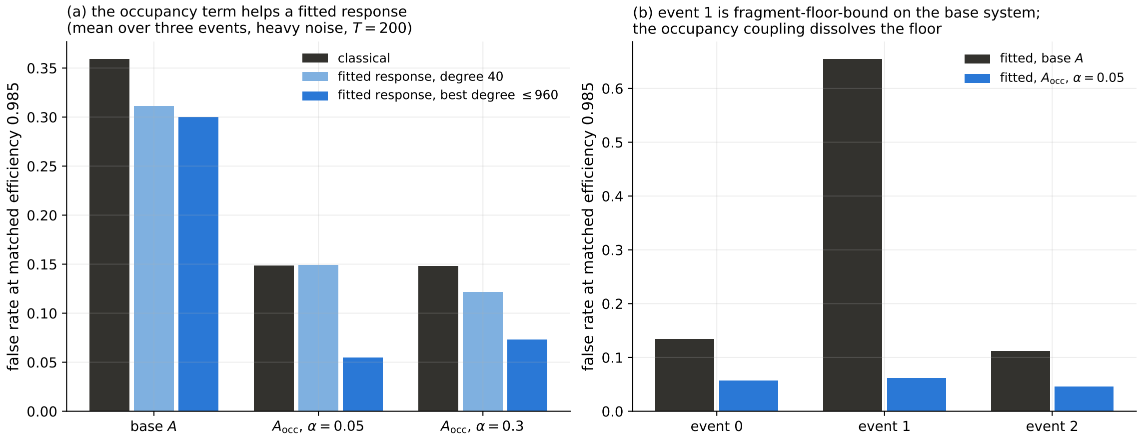


Figure 14: The occupancy term, refitted (heavy $T = 200$, degree-d responses fitted directly in coefficient space; identical machinery on every system). (a) False rate at matched efficiency 0.985, three-event means: the fitted response on the occupancy system at $\alpha = 0.05$ beats both the classical solver and the fitted response on the base system. (b) Per event: the event that is fragment-floor-bound on the base system (event 1) escapes the floor on the occupancy system — the co-hit coupling breaks the fragment/false-pair twin degeneracy.

pair line ($t = \pi/(s-1)$) does root the pile-up, and releases the isolated bulk, which re-enters at amplitude $|\cos(st/2)|\,\delta = 0.5$, on top of the true band, for a false rate of 0.995. The 1BQF's zero is spoken for. Covering both lines is degree two; covering the triple lines as well is degree four and up; and that, stated as a root budget, is in my opinion the cleanest justification for the whole QSVT programme that the toy has produced.

10 Limitations

I collect here, deliberately and in one place, everything that stands between the results above and a deployable algorithm. Each limitation comes with its measured size and, where we have one, its escape route.

10.1 Realizability of the fitted response

An earlier version of this study believed realizability was the bottleneck, and the false alarm is worth recording because the resolution strengthens the result. The binned optimum of eq. (18) has features of width ≈ 0.02 on a spectral span of ≈ 7; a Chebyshev series *reproducing that specific function* needs degree $d \approx 2W/h \approx 700$, and truncated series fitted to it at $d \leq 240$ performed terribly. But reproducing the binned optimum is the wrong problem. Fitting the objective of eq. (18) *directly in the degree-d coefficient space* — amplitudes are linear in the Chebyshev coefficients, so this is the same least-squares problem in $d{+}1$ unknowns — finds realisable responses that match the unlimited-degree optimum already at the production degree (figure 13d: 0.099 at $d = 40$ against the binned optimum's 0.096 at matched efficiency 0.98 on the base system). The measured discrimination of section 9 is therefore available at circuit-affordable degree; nothing about the polynomial family was ever in the way, only the indirect two-step fit.

A structural escape remains available on top: the same no-giant-component fact that made the spectrum measurable makes the problem block-diagonal. Solving per connected component (a classical $\mathcal{O}(n)$ preprocessing step) replaces one $4T^2$-dimensional filter problem by hundreds of ≤ 17-dimensional ones, on which a degree-≤ 17 interpolating polynomial reproduces any response exactly, and as a bonus the register shrinks to $\lceil \log_2 17 \rceil + 2$ qubits and the success probability per cluster becomes $\mathcal{O}(1)$. The cost is honesty about what remains quantum: per-cluster solving helps the classical solver identically, and the quantum claim reduces to the resource profile of section 4.3 on whatever cluster sizes survive preprocessing.

10.2 Generalization of the fit

The response fitted on one event transfers poorly to its statistical siblings (0.63–0.99 false rate on cross-application, against 0.04–0.10 in-sample). The decomposition is instructive. The root positions are universal ($s-1$ and $s \pm \sqrt{2}$ are motif properties, not event properties), but the fitted pass positions chase each event's particular contaminated-true modes, which shift with the noise realisation. A production design must therefore be a universal response: roots pinned to the motif lines, passes wide enough to cover the contaminated-true spread measured across an event ensemble, fitted jointly rather than per event. The per-event fit of section 9 then plays the role its construction suggests, the measured ceiling that a universal design approaches. Building that universal response, and quantifying the ceiling gap, is the programme's clear next step.

10.3 Trotterization on hardware

Everything above uses exact evolution. On hardware, the 1BQF's product-formula circuit is exact only in the uniform-coupling case; the moment coupling weights differ (erf-weighted kernels, fractional fork strengths) the naive per-pair formula scatters the readout by amounts that dominate the physics (section 8). The structured-synthesis route (edge-coloured pair ordering, developed in the programme's DSS study, which also reaches 1BQF width for the comb via generalized QSP over e^{-iAt}) restores exactness at matched cost and is, on present evidence, mandatory for any weighted Hamiltonian on hardware.

10.4 Success probability and readout

The filtered state must still be post-selected and read out, and it is worth being explicit about what that costs, because it is the largest single overhead in the whole scheme (figure 15). The LCU post-selection succeeds when the index and dilation registers both return zero, which happens with probability P_{anc} of eq. (14); physically, P_{anc} is the fraction of the input state's norm that survives the filter. Because the uniform input spreads over all $4T^2$ segments while only $\mathcal{O}(T)$ of them carry surviving amplitude, P_{anc} falls as $1/T$: measured $\approx 0.44/T$ for the comb and estimated $\approx 0.25/T$ for the 1BQF, i.e. thousandths at $T = 1000$. Left alone this would mean thousands of repetitions per useful sample. Amplitude amplification (the standard Grover-type trick) reduces the repetition count from $1/P_{\text{anc}}$ to $1/\sqrt{P_{\text{anc}}}$, restoring a $\sqrt{T}$ scaling of total walk calls per useful sample, and active-set readout (sampling until the $\mathcal{O}(T)$ surviving segments are each seen) gives the measured $d \cdot T^{1.5} \log T$ walk-call budget against the classical $d \cdot T^2$ flop budget: a square-root-class separation for that readout task, and no more than that.

10.5 Classical simulability of the response family

Every response function in this paper can be applied classically by the Chebyshev recursion in $\mathcal{O}(d \cdot \text{nnz}(A))$ flops, seconds at $T = 1000$. The discrimination physics (combs, roots, floors) is solver-independent mathematics of the segment Hamiltonian, and this paper's contribution to it stands whether or not a quantum computer is ever involved. The quantum claim is confined to the resource profile: width logarithmic in problem size, walk calls $d\sqrt{T}$ against classical dT^2 for the readout task above, and a hardware-native route to the same amplitudes. Readers allergic to quantum advantage claims may read the whole paper as classical spectral-filter engineering with a quantum implementation appendix; nothing in the physics sections would change.

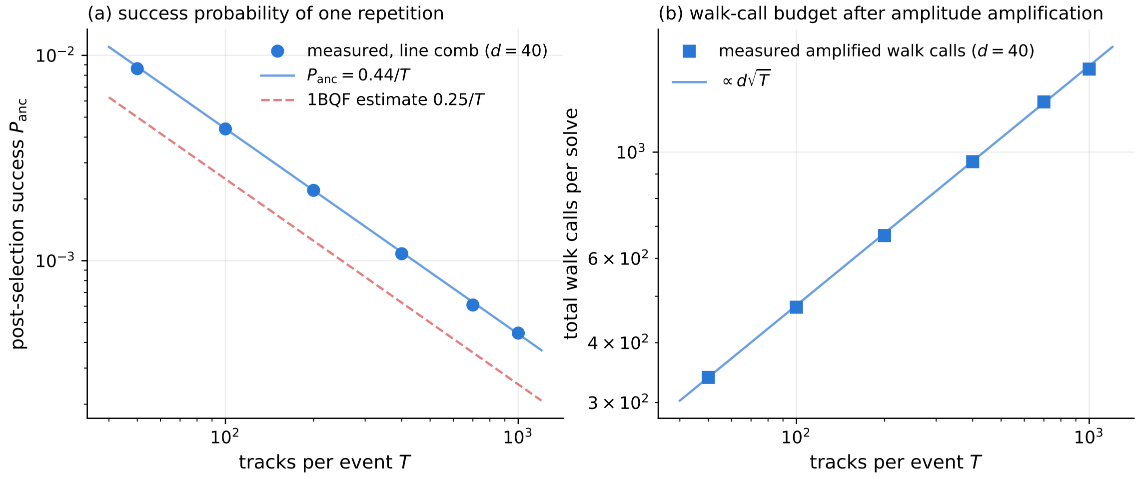


Figure 15: The readout budget, measured on the shared events at the production degree $d = 40$. (a) Post-selection success probability per repetition: the measured comb points follow $0.44/T$; the 1BQF estimate $0.25/T$ is shown for comparison. (b) Total walk calls per solve after amplitude amplification, following the $d\sqrt{T}$ scaling.

10.6 The toy's geometry, and what a real detector changes

Finally, the largest caveat of all: the four-line comb exists because this toy pins every true track to five hits. Real VELO tracks have variable length, and a companion study on real Run-3 event geometry found the true spectrum smeared into a band, the P_4 comb keeping only 40% of true segments, and a short-fragment floor ($m \leq 2$ tracks indistinguishable from noise at the segment level) capping all segment-level solvers, while the fork transfer and the band-plus-notch redesign behaved as the toy predicts. Figure 16 shows the measured amplitude pile-up on the real geometry, side by side for the three solver designs: where the toy's amplitude atlas (figure 1) has discrete levels, the real geometry smears the true population into a continuum, the fragment population ($m \leq 2$, orange) sits inside the false bulk for every solver, and the redesigned band-plus-notch response reproduces the toy's qualitative separation on the findable ($m \geq 3$) population. The honest statement of scope is that this paper's method is response engineering on whatever spectral structure the geometry provides, demonstrated exhaustively on a geometry that provides a lot of it; the measurement loop of section 9 (measure the pile-up, fit the response, locate the floor) is geometry-agnostic even where the specific comb is not.

11 Conclusion

This paper set out to finish a thought that the one-bit quantum filter started. The 1BQF's insight was that track finding does not need a linear-system solution, only a spectral separation, and that one cosine with one zero buys the largest separation in the problem at almost no quantum cost. Promoting that cosine to the full QSVT polynomial family converts algorithm design into response design, and response design turns out to be a measurement-driven loop:

1. Read the spectrum the problem gives you. On clean events the detector geometry hands you four exact eigenvalue lines, and a fixed forty-degree comb pinned to them converts the segment selection at $T = 1000$ tracks from a 20.5% false rate (the classical solver at its fixed threshold) to 0.9% (the comb at the same threshold, at 92% segment efficiency): a factor twenty fewer fakes in the selected sample.

2. When the problem changes, measure where the false weight went, and put roots there. At the heavy-noise operating point ($\sigma_{\text{res}} = 20\,\mu\text{m}$, 1% hit drop, $T = 200$) the false weight piles

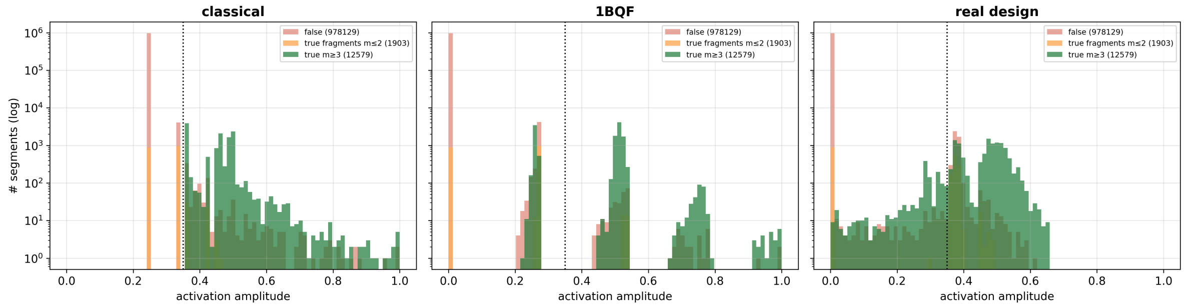


Figure 16: The real-geometry counterpart of the toy's amplitude pile-up: activation-amplitude spectra on real Run-3 VELO event geometry, for the classical solver, the 1BQF, and the redesigned band-plus-notch response. Pink: false segments; orange: true fragments ($m \leq 2$); green: findable true segments ($m \geq 3$); the dotted line is the fixed threshold. Variable track length smears the discrete toy levels into bands, and the fragment population sits inside the false bulk for every solver — the real-geometry form of the fragment floor.

up on the pair and triple motif lines, and the response refitted to that measurement reads a false rate of 0.061 at matched segment efficiency 0.97 where the classical solver on the same events reads 0.124 — half the fakes at identical efficiency, on every event tested, at every efficiency below the fragment floor.

3. Judge every Hamiltonian modification together with the response fitted to it, never separately. Both Denby–Peterson terms destroy the degree-one filter, and both help the refitted polynomial: the fork system fits to 0.046 at matched efficiency 0.97 (against 0.061 without the fork), and the occupancy system at small coupling $\alpha = 0.05$ fits to 0.039 — the best number in the study, produced by the same term whose coupling collapses the 1BQF's separation (AUC 0.996 → 0.634). Operator and response are one design, not two.

4. Know what the spectrum cannot see — and know that the operator can change what that is. Spectral twins bound every function of λ on a fixed Hamiltonian, classical scoring included; at heavy noise the true fragments of broken tracks are twins of the false pairs, and on the base and fork systems they floor every solver at segment efficiency ≈ 0.98: pushing to 0.99 floods the selection back to a ≈ 0.65 false rate for everything. But the floor is a degeneracy of the operator, not of the event: the occupancy coupling embeds each motif in its hit-sharing neighbourhood, breaks the twin isomorphism, and the one event that was floor-bound at 0.654 on the base system fits to 0.062 on the occupancy system at the same matched efficiency.

The open problems are, I think, unusually crisp for this field: a universal (event-ensemble) response to replace the per-event fit, which is the gap between the measured ceilings above and a deployed algorithm; the same universality question for the floor-breaking occupancy result; per-cluster realization as the structural alternative to global filtering; edge-coloured synthesis for weighted couplings on hardware; and the transplant of the whole loop onto real detector geometry, where the first measurements say the method survives even though the comb does not. Every cloud has a silver lining: the same noise that broke the clean-event comb handed us a false population parked on nameable eigenvalue lines, and a filter family with exactly enough roots to meet it.

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